

DEFINISI FORMAL DFA

Deterministic finite automata (DFA) $\rightarrow \mathbf{M} = (\mathbf{Q}, \Sigma, \delta, q_0, F)$,

dimana :

Q : himpunan state/kedudukan

Σ : himpunan simbol input

δ : fungsi transisi, dimana $\delta \in Q \times \Sigma \rightarrow Q$

q_0 : State awal (initial state)

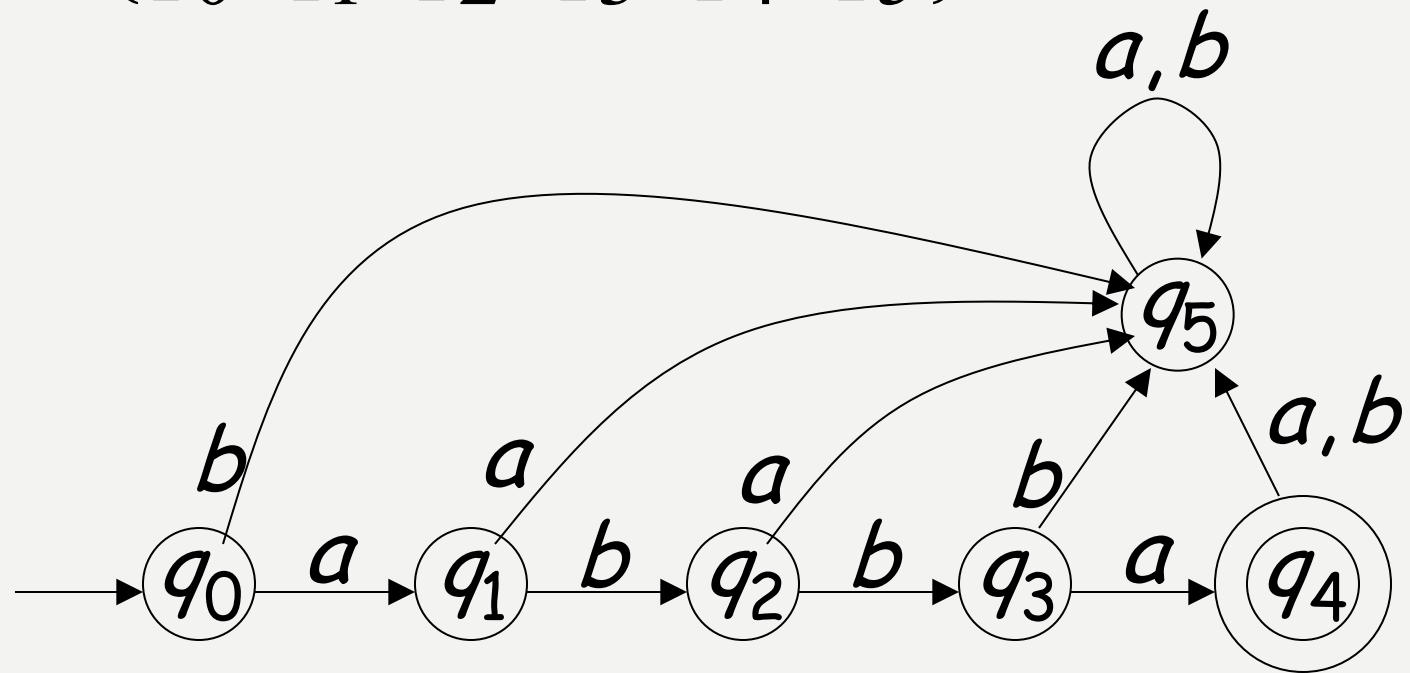
F : himpunan state akhir (Final State)

Language $\rightarrow L(M) : \{x \mid \delta(q_0, x) \in F\}$

HIMPUNAN STATE Q

Contoh

$$Q = \{q_0, q_1, q_2, q_3, q_4, q_5\}$$

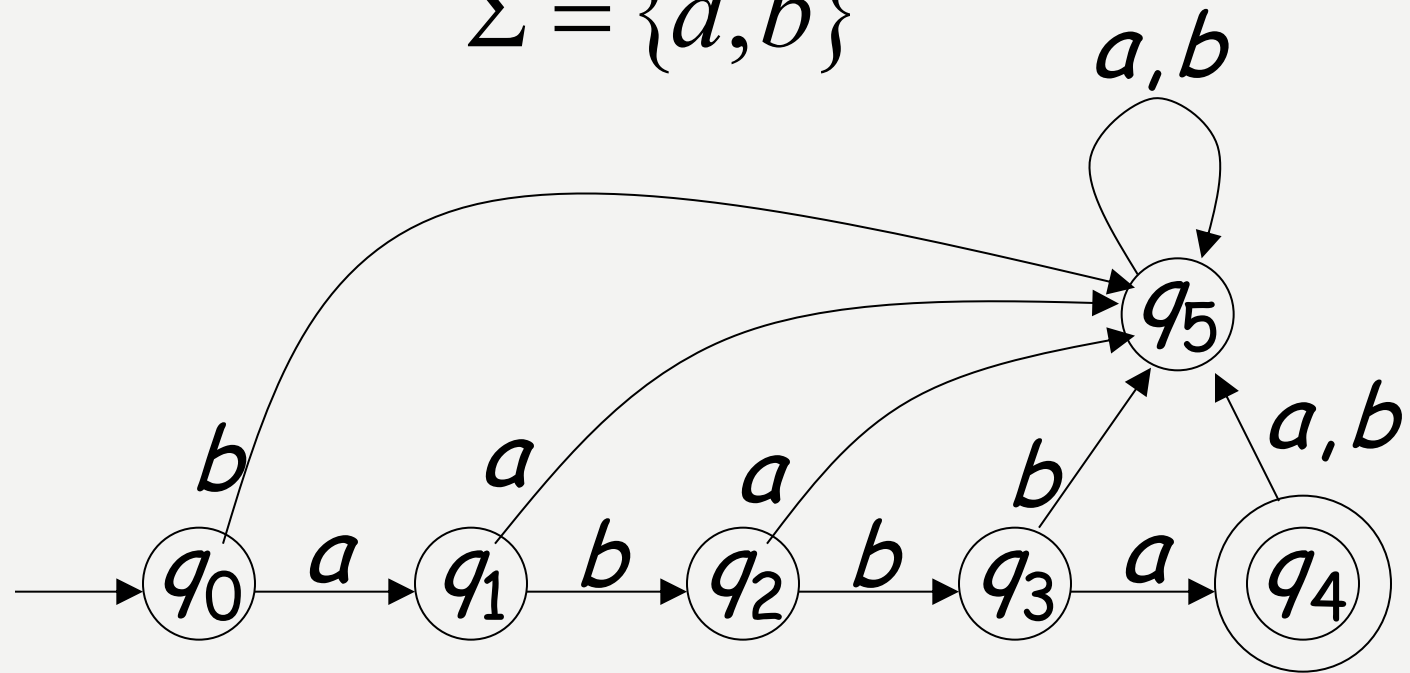


INPUT ALPHABET Σ

$\varepsilon \notin \Sigma$: input alphabet tidak pernah berisi *empty string* / ε

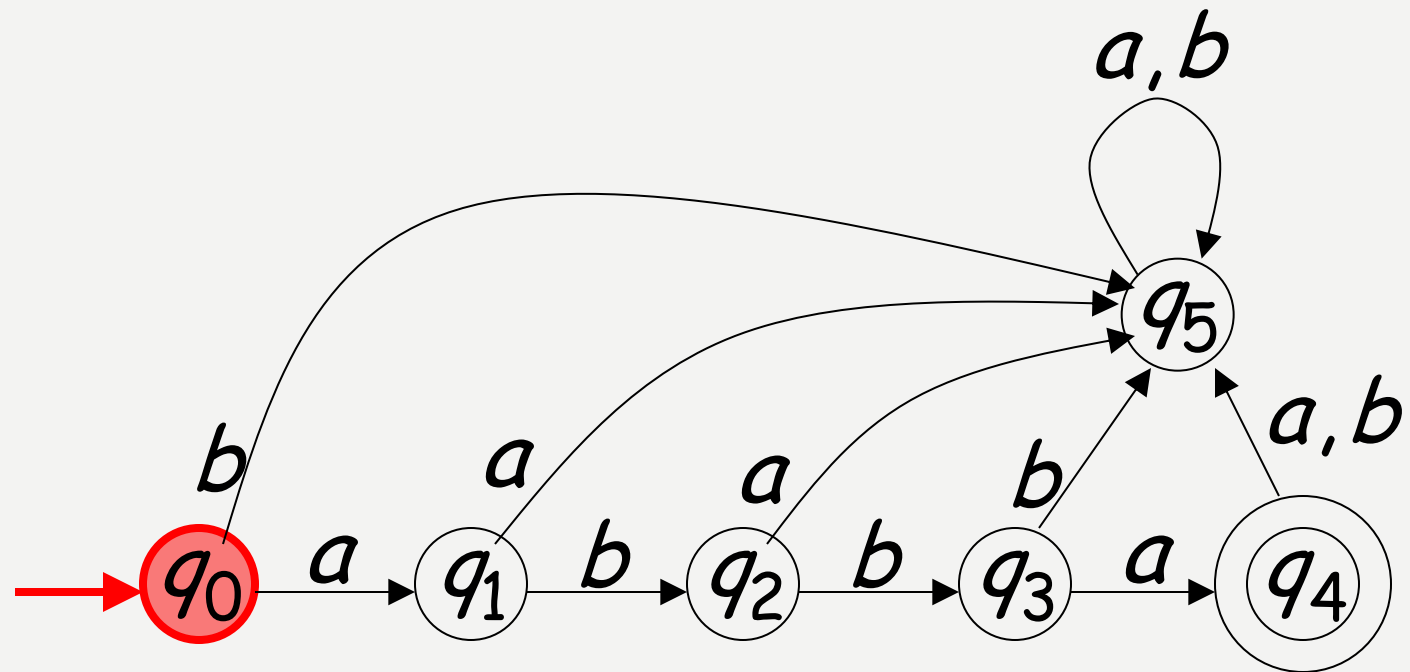
Contoh

$$\Sigma = \{a, b\}$$



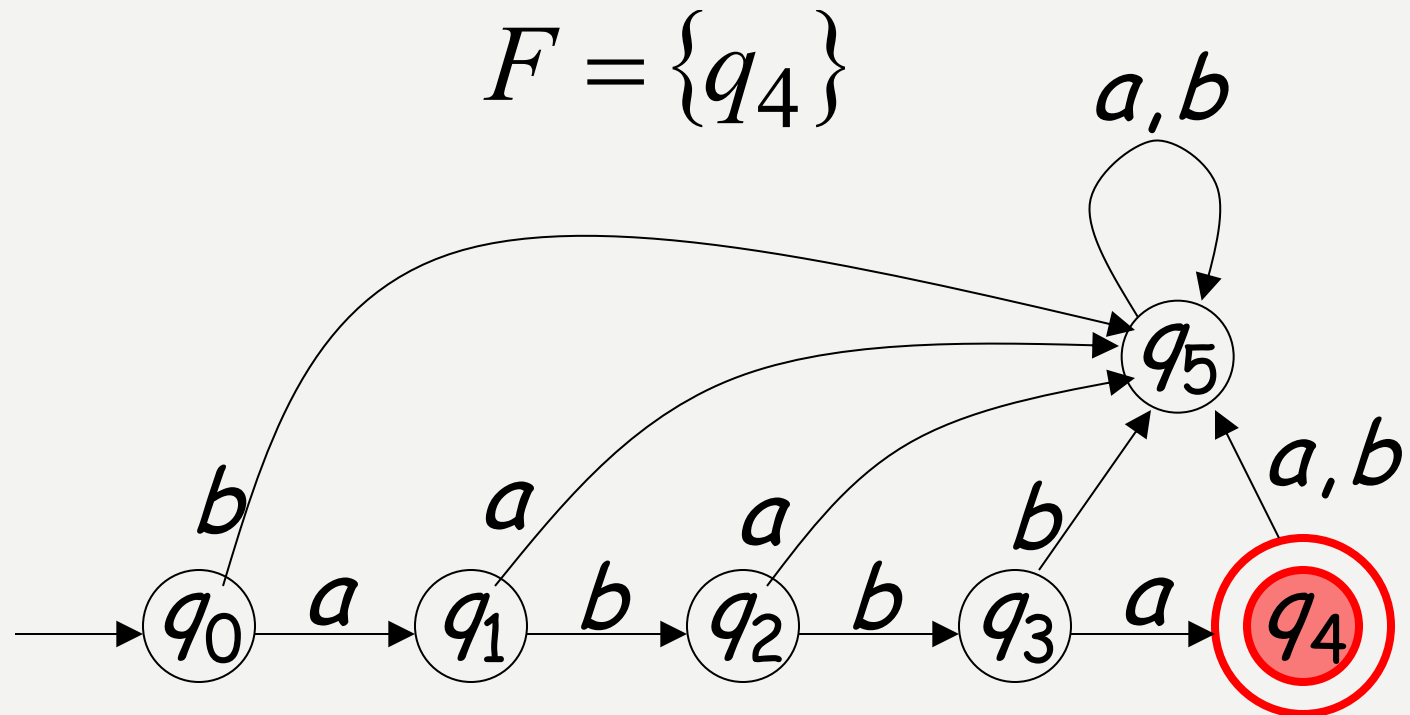
INITIAL STATE q_0

Contoh



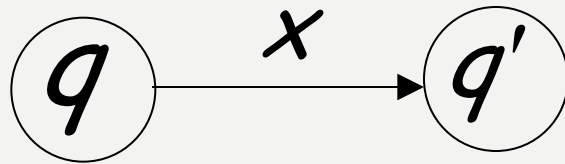
HIMPUNAN FINAL STATES $F \subseteq Q$

Contoh



FUNGSI TRANSISI $\delta: Q \times \Sigma \rightarrow Q$

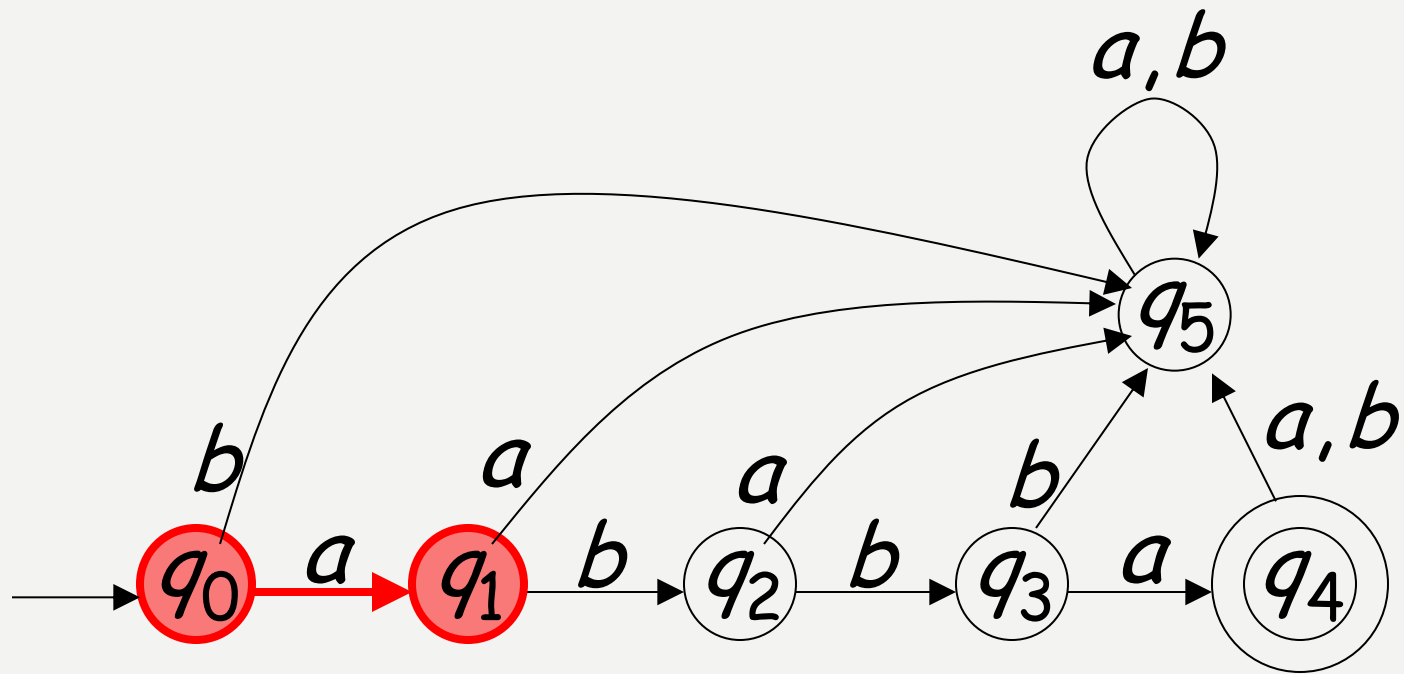
$$\delta(q, x) = q'$$



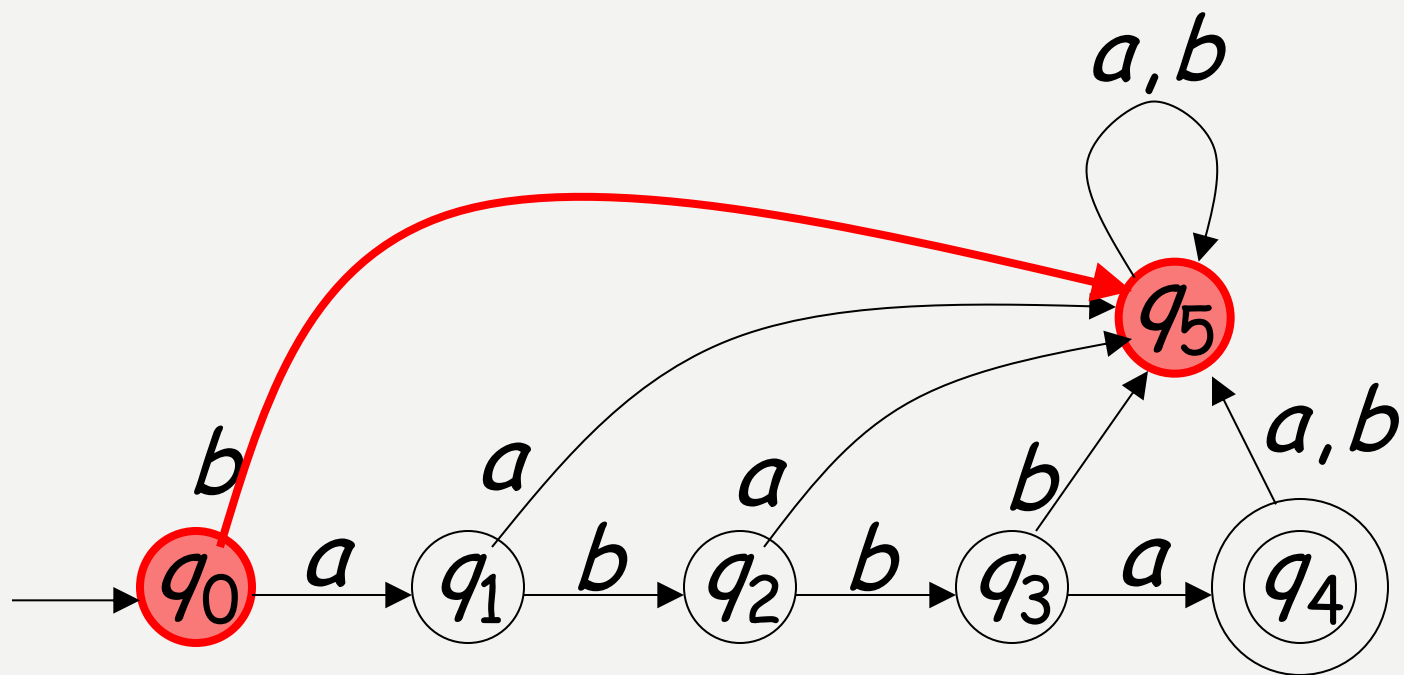
Menggambarkan hasil transisi dari state q dengan input x

Contoh

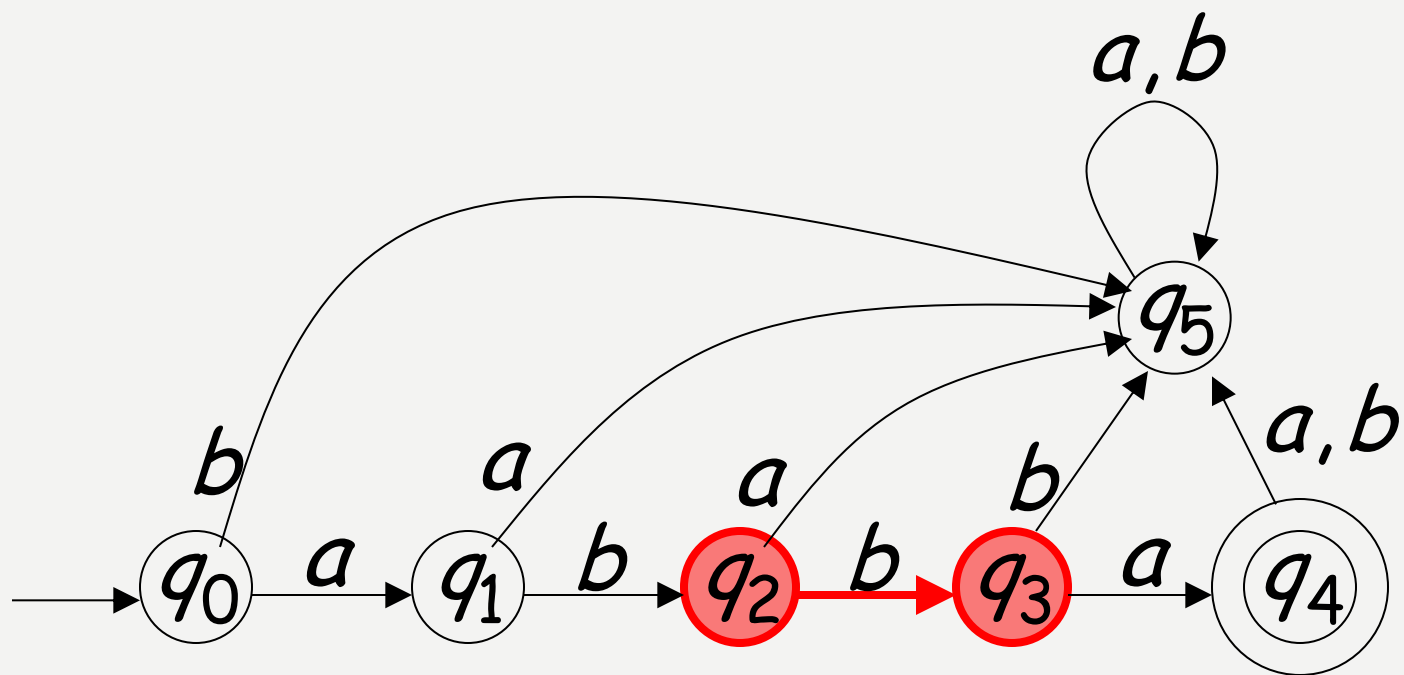
$$\delta(q_0, a) = q_1$$



$$\delta(q_0, b) = q_5$$



$$\delta(q_2, b) = q_3$$

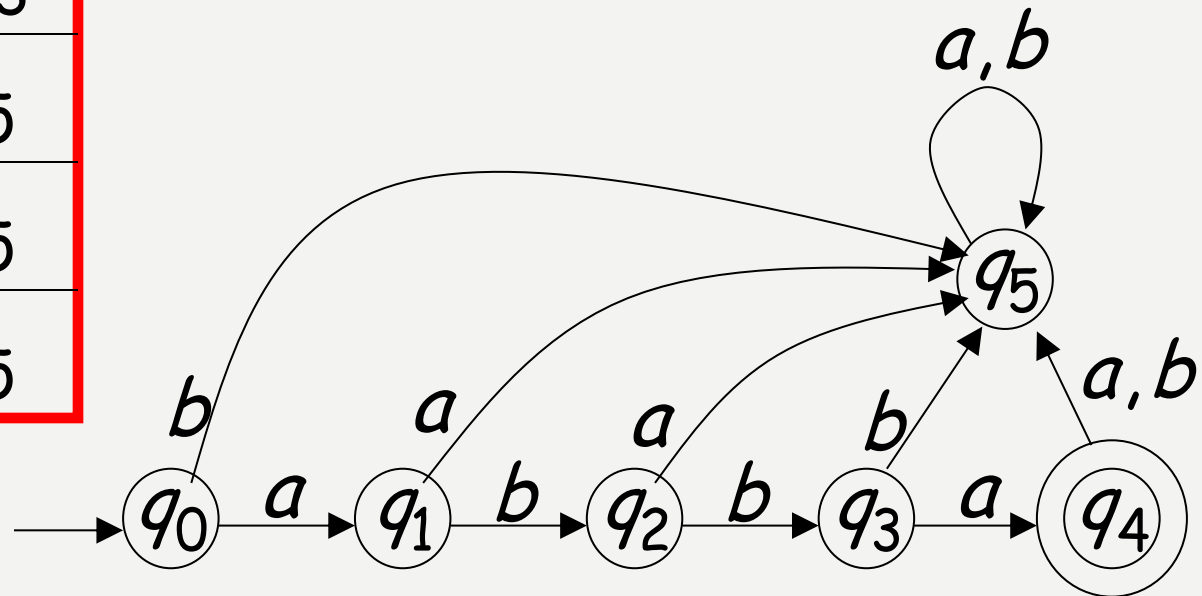


TABEL TRANSISI δ

symbols

states

δ	a	b
q_0	q_1	q_5
q_1	q_5	q_2
q_2	q_5	q_3
q_3	q_4	q_5
q_4	q_5	q_5
q_5	q_5	q_5



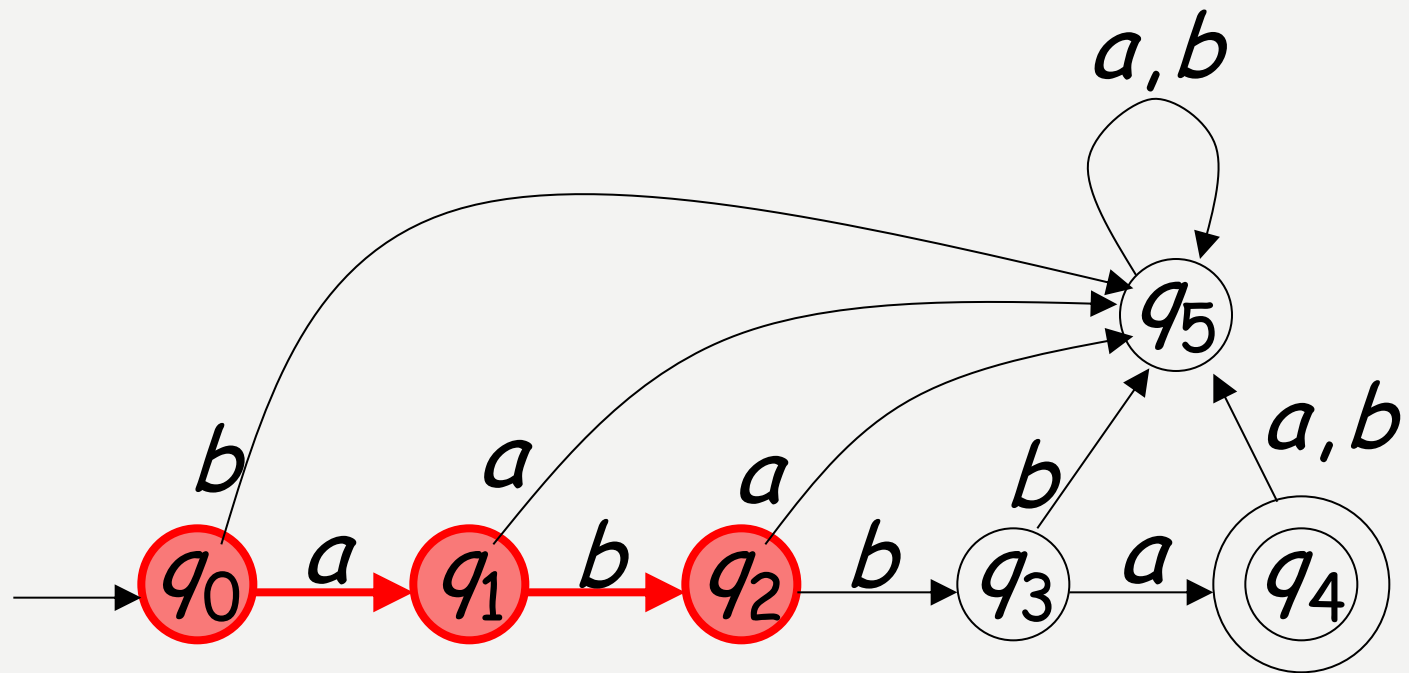
FUNGSI TRANSISI DENGAN INPUT STRING

$$\delta^* : Q \times \Sigma^* \rightarrow Q$$

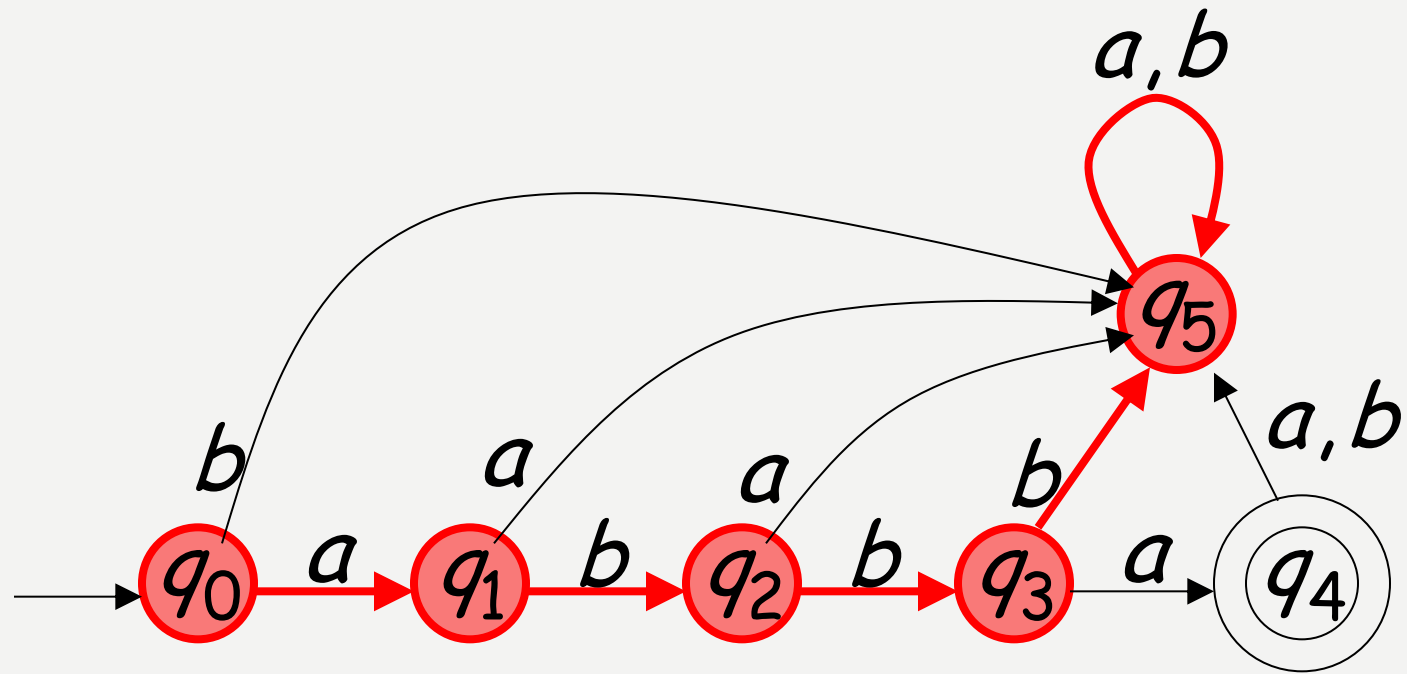
$$\delta^*(q, w) = q'$$

Menggambarkan state yang diperoleh dengan input string w dari state q

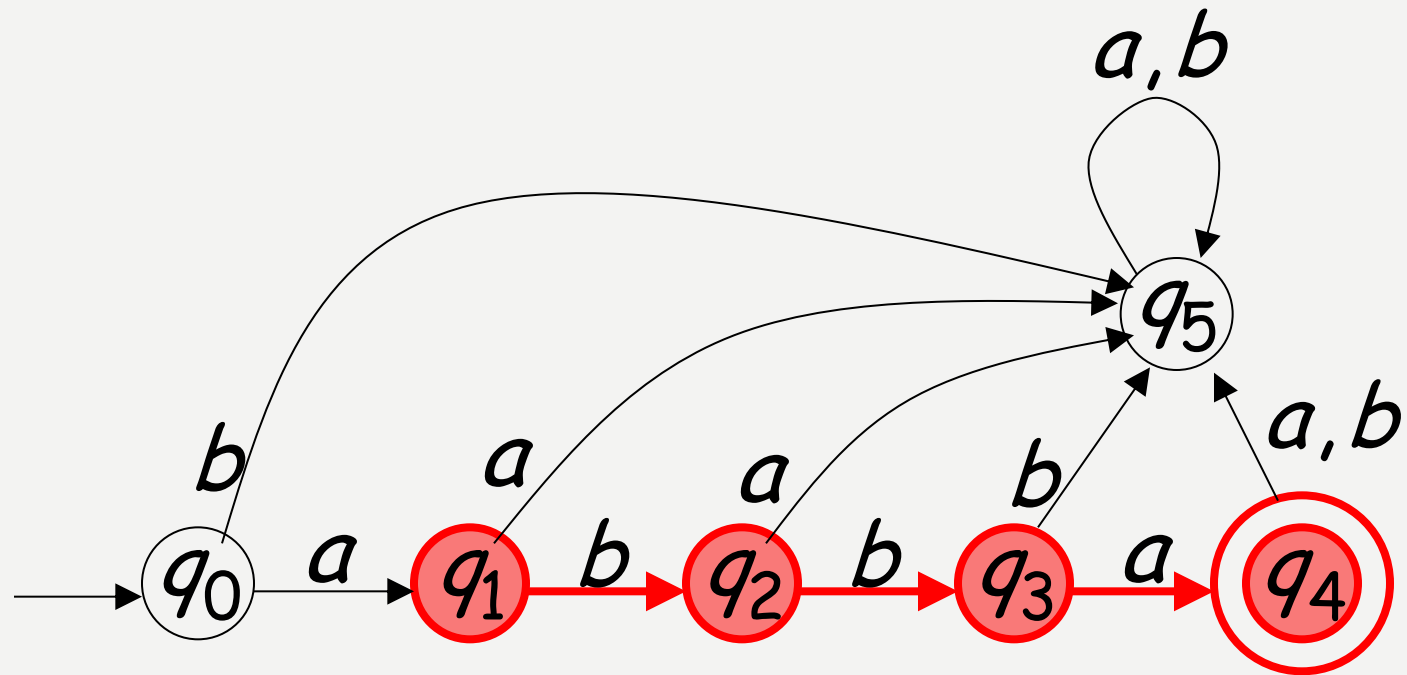
Contoh: $\delta^*(q_0, ab) = q_2$



$$\delta^*(q_0, abbbaa) = q_5$$

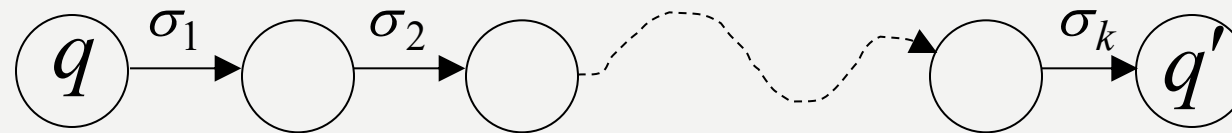


$$\delta^*(q_1, bba) = q_4$$



Secara umum: $\delta^*(q, w) = q'$

$$w = \sigma_1 \sigma_2 \cdots \sigma_k$$



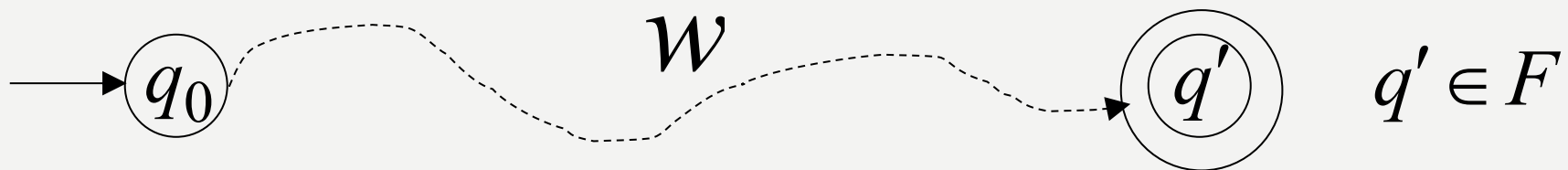
BAHASA YANG DITERIMA DFA

Bahasa yang diterima oleh DFA M

Dinotasikan sebagai $L(M)$ dan berisi semua string yang diterima oleh M

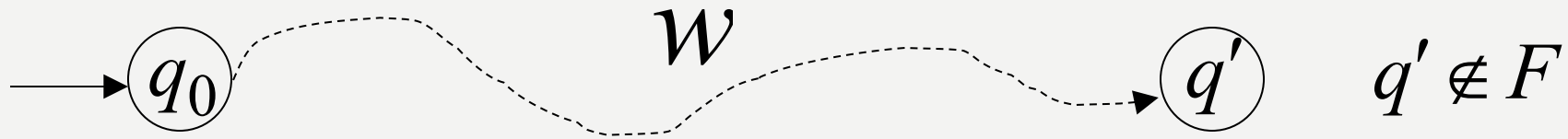
Untuk DFA $M = (Q, \Sigma, \delta, q_0, F)$

Bahasa yang diterima $L(M) = \{w \in \Sigma^* : \delta^*(q_0, w) \in F\}$

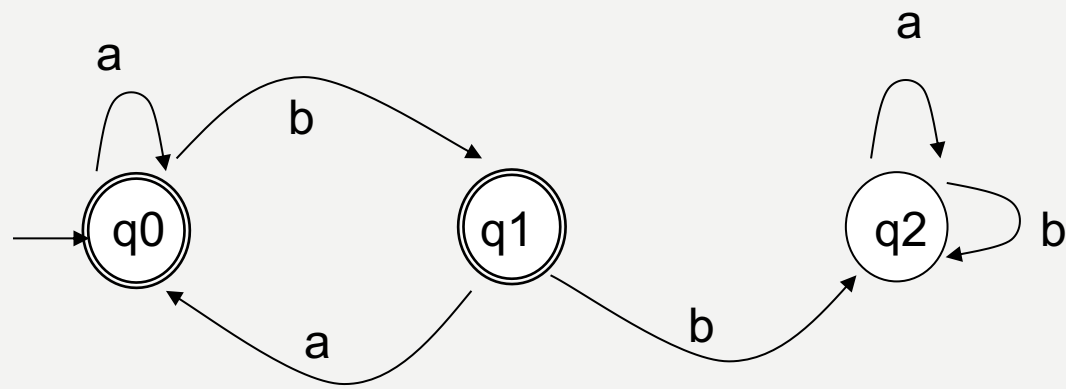


Bahasa yang ditolak

$$\overline{L(M)} = \{w \in \Sigma^* : \delta^*(q_0, w) \notin F\}$$



CONTOH



DFA :

$Q = \{q_0, q_1, q_2\}$

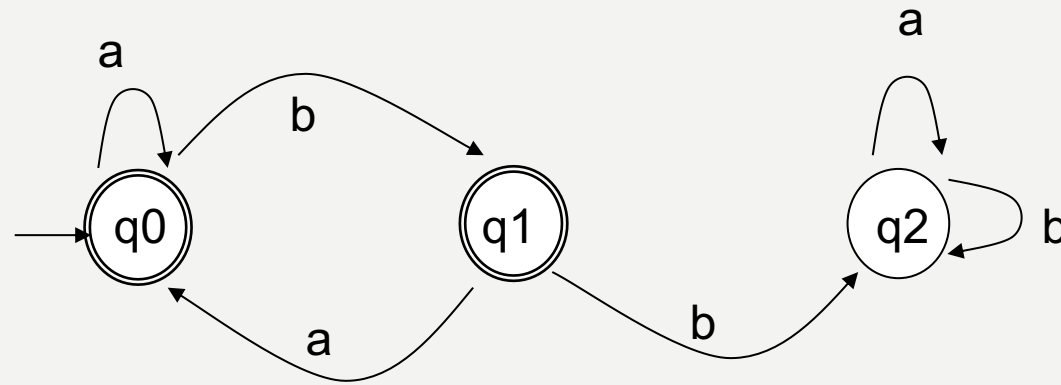
$\Sigma = \{a, b\}$

$S = q_0$

$F = \{q_0, q_1\}$

δ diberikan dalam tabel berikut :

δ	a	b
q0	q0	q1
q1	q0	q2
q2	q2	q2



Jawab :

$\delta(q_0, abababaa) \Rightarrow \delta(q_0, bababaa) \Rightarrow \delta(q_1, ababaa) \Rightarrow$
 $\delta(q_0, babaa) \Rightarrow \delta(q_1, abaa) \Rightarrow \delta(q_0, baa) \Rightarrow \delta(q_1, aa) \Rightarrow$
 $\delta(q_0, a) \Rightarrow q_0$

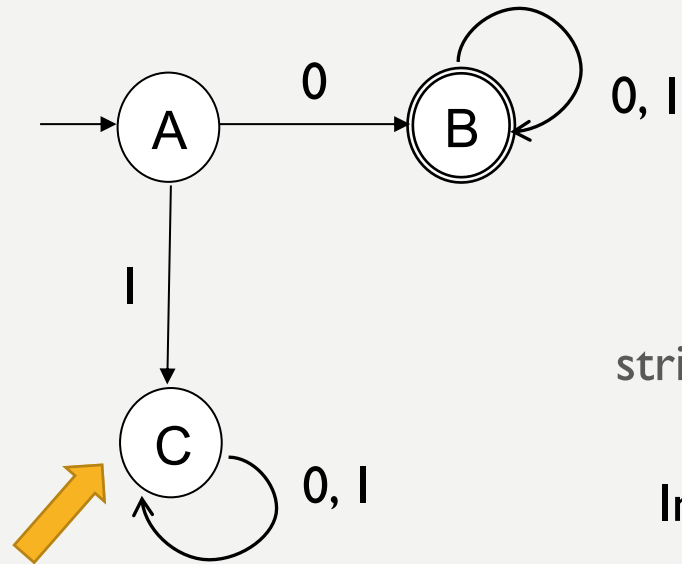
Tracing berakhir di q_0 (state AKHIR) $\Rightarrow$ kalimat abababaa diterima

$L(M) = \{abababaa, aaaabab, aabababa, \dots\}$

MENYUSUN DFA

CONTOH 1

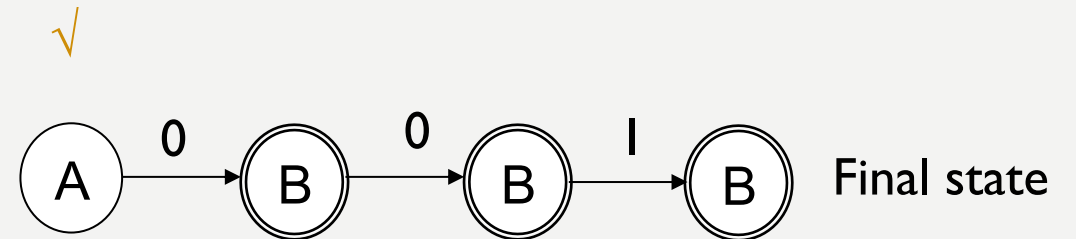
- LI = himpunan semua string yang dimulai dengan '0'
 $= \{0, 00, 01, 000, 010, 011, 0000, \dots\}$



Dead State /
Trap State

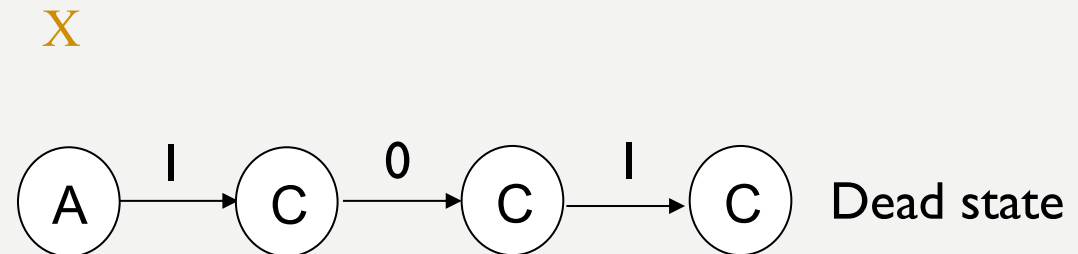
string : 001

Initial state



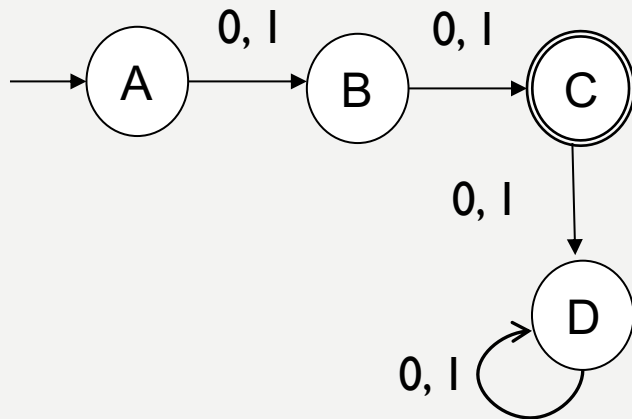
string : 101

Initial state

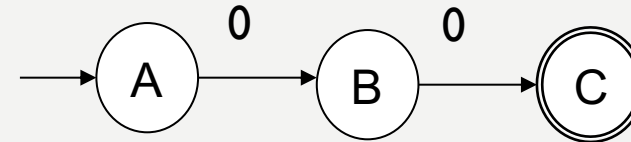


CONTOH 2

- Susun DFA yang menerima himpunan semua string $\{0, 1\}$ yang panjangnya 2.
- $\Sigma = \{0, 1\}$
- $L = \{00, 01, 10, 11\}$

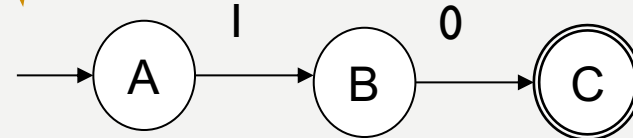


• String : 00 ✓



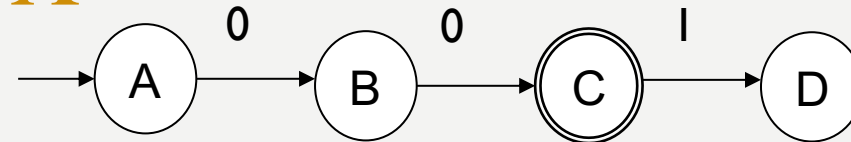
Final state

• String : 10 ✓



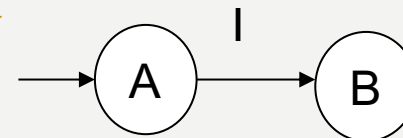
Final state

• String : 001 X



Dead state

• String : 1 X



Dead state



THANK YOU